

Activation functions:

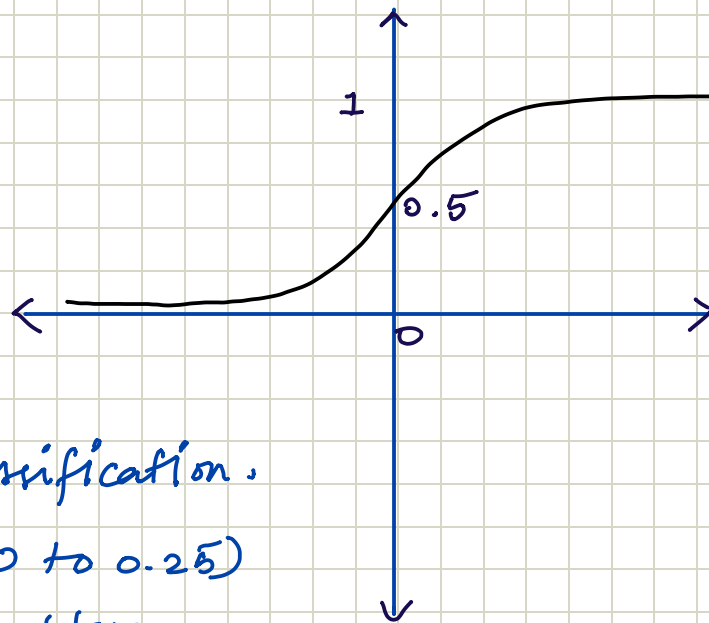
It is a mathematical function applied to the output of a neuron. It introduces non linearity to the model, allowing it to learn and represent complex pattern in the data.

$$\textcircled{*} \quad e^{-\infty} = 0 \quad e^0 = 1 \\ e^{\infty} = \infty$$

Types of Activation Functions:

(i) Sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Range: (0, 1)

- * Mainly used for Binary classification.
- * The derivative ranges from (0 to 0.25)
Leads to vanishing gradient problem.

Pros:

- * Smooth Gradient
- * (0 to 1) $\rightarrow$ Normalizing the o/p of the neuron
- * Clear Predictions 0 or 1

Cons:

- * prone to gradient Problem
- * function is not zero centered. (gradients become very small)
- * power operation so time consuming.

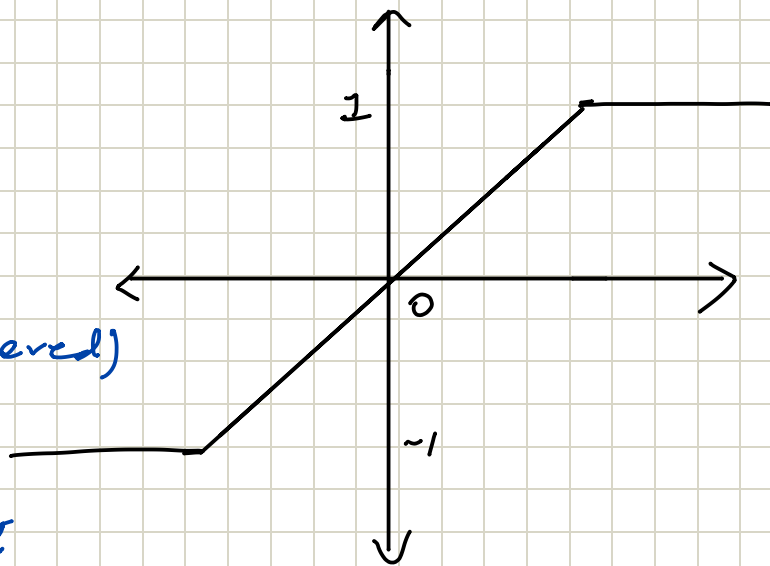
2.) tanh function:

tanh is a hyperbolic tangent function, shifted version of sigmoid.

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Ranges: $(-1 \text{ to } 1)$ (zero centered)

Similar to sigmoid, the only difference is the output range.



Sigmoid $\rightarrow$ Output Layer (Binary classification)

tanh $\rightarrow$ hidden Layers.

(3.) Relu (Rectified Linear Unit)

$$f(x) = \max(0, x)$$

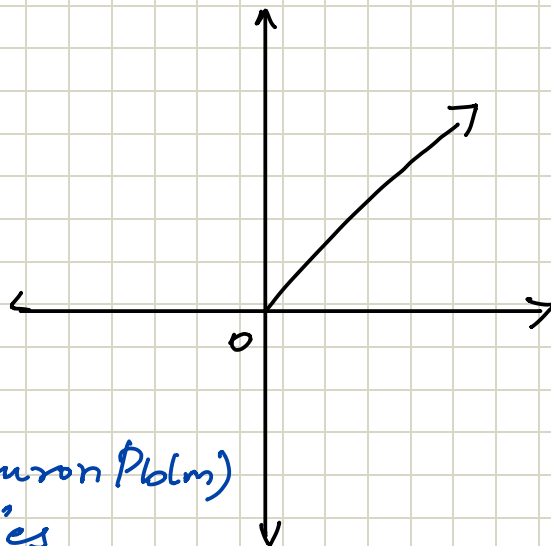
Range: 0 to ∞

Pros: * Computation is faster single max function.

* Input is (+ve), no gradient saturation.

Cons: When input is (-ve), Relu dies (dead neuron Pblm)

(i.e) $W_{old} \approx W_{new}$ (vanishing gradient Problem)
not zero centered.



(4.) Leaky Relu function:

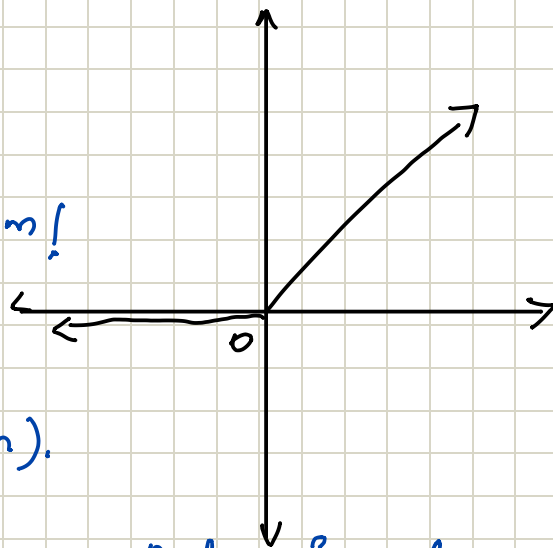
$$f(x) = \max(0.01x, x)$$

Range: $(-\infty \text{ to } \infty)$

This solves the dead neuron problem!

Leaky Relu has all the pros of Relu + (no dead neuron problem).

It is not fully proved that Leaky Relu is always better than Relu.



Parameteric Relu: $f(x) = \max(\alpha x, x)$

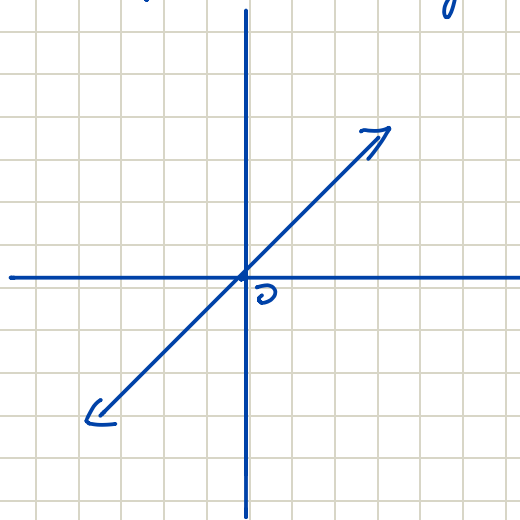
Alpha can be learned from back propagation!

(5.) Linear Activation fn:

$$y = x$$

Range: $-\infty \text{ to } \infty$

This Linear Activation combined with non linear activation fn to enhance neural n/w learnings & predictive capabilities.



6.) Softmax function:

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

* Used for multiclass classification problems. Converts neuron outputs to probabilities.

* Non Linear Activation function & each class is assigned a probability helps in identify which class it belongs to.

Note: Sum of Probability = 1

7.) Softplus function:

$$\sigma(x) = \log(1 + e^x)$$

This ensures the output is always positive & differentiable at all points which is advantage over traditional ReLU.

* Smooth, Continuous function (avoids sharp discontinuities of ReLU).

Impact of Activation functions on Model Performance

	Classification	Regression
* Convergence Speed	B $\rightarrow$ ReLU + Sigmoid	ReLU Linear
* Gradient flow	M $\rightarrow$ ReLU + Softmax	
* Model Complexity.		

8.) Swish (a.k.a Sigmoid Linear Unit (SILU))

$$f(x) = x \cdot \text{Sigmoid}(\beta \cdot x)$$

$$\text{Sigmoid} = \frac{1}{1 + e^{-x}}$$

β - constant or learning parameter

Swish is like ReLU, but smoother.

If x is large $\Rightarrow$ output is almost x .

If x is small or negative $\Rightarrow$ output is small but not zero.

⊛ So instead of cutting the negative values completely like ReLU, it just slows them down.

Pros:

- * Smooth Gradient
- * No dead neuron problem
- * used in efficientNet / vision transformers.

Cons:

- * Slower Than ReLU(max).
- * Non zero centered
- * hard to interpret.
- * Hyperparameter Sensitivity ($\beta = \pm$) often.

When to use: When you already using ReLU but if you want small boost in performance

(9.) Gaussian Error Linear Unit (GELU)

$$f(x) = x \cdot \Phi(x)$$

$\Phi(x)$ is The probabilistic function determines the probability of x is positive. (how usefulness determination)

GELU acts as a soft, probabilistic version of ReLU.

Intuition:

If x is large $\Rightarrow$ it almost passes fully ($\approx x$)

If x is slightly +ve } $\Rightarrow$ half passes ($0.5x$)
or almost zero }

If x is negative $\Rightarrow 0$

Instead of hard "on/off" in ReLU, GELU smoothly fades out negative numbers.

Pros:

- * Smoother Learning
- * Better Gradient flow
- * Great for text (Transformers)
- * Probabilistic Gating

Cons:

- * slower
- * Computational heavier
- * Non zero centered
- * hard to interpret.

Comparison Relu, Swish & GReLU.

Activation	Type	Smooth	Used	Speed
Relu	Simple Cutoff	No	all CNNs	fast
SiLU	Smooth self gating	Yes	EfficientNet	medium
GReLU	Prob. Smooth	Very Yes	Transformers	slow.